UNIT – IV: Packages and Java Library

Introduction, Defining Package, Importing Packages and Classes into Programs, Path and Class Path, Access Control. Exception Handling: Introduction, Hierarchy of Standard Exception Classes, Keywords throws and throw, try, catch, and finally Blocks, Multiple Catch Clauses, Class Throwable, Unchecked Exceptions, Checked Exceptions.

//java program to create a thread

```
import java.lang.*;
public class ThreadDemo
{
	public static void main(String args[])
	{
		Thread obj=Thread.currentThread();
		System.out.println("The thread is"+obj);
		obj.setName("Open elective");
		System.out.println("The thread is"+obj);
		obj.setPriority(7);
		System.out.println("The thread is"+obj);
	}
}
/*output
D:\OpenElectiveOOP>java ThreadDemo
The thread isThread[main,5,main]
The thread isThread[Open elective,5,main]
The thread isThread[Open elective,7,main]*/
```

//java program to demonstrate the usage of throw statement

```
import java.lang.*;
public class ThrowDemo
{
	public static void main(String args[])
	{
		try
		{
			throw new ArithmeticException("You are trying to perform division
by zero");
		}
		catch(ArithmeticException e)
		{
```

```
                        System.out.println(e.getMessage());
                        e.printStackTrace();
                }
        }
}
```

//java program to demonstarte the throws statement

```
import java.lang.*;
import java.util.*;
class Test
{
        Test() throws ArithmeticException,ArrayIndexOutOfBoundsException
        {
                int a;
                Scanner s=new Scanner(System.in);
                System.out.println("Enter the value of a");
                a=s.nextInt();
                int b=10/a;
                System.out.println("The value of b is"+b);
                if(a==1)
                {
                        a=a/(a-a);
                }
                if(a==2)
                {
                        int c[]=new int[-10];
                        c[45]=99;
                }
        }
}
public class ThrowsDemo
{
        public static void main(String args[])
        {
                try
                {
                        Test t=new Test();
                }
                catch(Exception e)
                {
```

```
                    System.out.println(e.getMessage());
                    e.printStackTrace();
            }
        }
}

            .


import java.lang.*;
class Reservation extends Thread
{
        int available=1;

        int wanted;
        Reservation(int i)
        {
                wanted =i;
        }
        public void run()
        {
                synchronized(this)
                {
                        System.out.println("Available Berths"+available);
                        if(available>=wanted)
                        {
                                String name=Thread.currentThread().getName();
                                System.out.println("Berth allocated to"+name);
                                try
                                {
                                        Thread.sleep(1000);
                                        available=available-wanted;
                                }
                                catch(InterruptedException e)
                                {
                                        System.out.println(e.toString());
                                }
                        }
                        else
                        {
                                System.out.println("No Berths");
```

```
				}
			}
		}
}
public class SynchronizedDemo
{
		public static void main(String args[])
		{
			Reservation r=new Reservation(1);
			Thread t1=new Thread(r,"Prakash");
			Thread t2=new Thread(r,"Srinivas");
			t1.start();
			t2.start();
		}
}
```

//java program to create thread by extending thread class

```
import java.lang.*;
class Thread1 extends Thread
{
		public void run()
		{
			try
			{
				for(int i=0;i<5;i++)
				{
					System.out.println("Good Morning");
					Thread.sleep(1000);
				}
			}
			catch(InterruptedException e)
			{
				System.out.println(e.getMessage());
			}
			System.out.println("Thread1 exit");
		}
}
class Thread2 extends Thread
{
		public void run()
```

```
	{
		try
		{
			for(int i=0;i<5;i++)
			{
				System.out.println("Hello");
				Thread.sleep(2000);
			}
		}
		catch(InterruptedException e)
		{
			System.out.println(e.getMessage());
		}
		System.out.println("Thread2 exit");
	}
}
class Thread3 extends Thread
{
	public void run()
	{
		try
		{
			for(int i=0;i<5;i++)
			{
				System.out.println("Welcome");
				Thread.sleep(3000);
			}
		}
		catch(InterruptedException e)
		{
			System.out.println(e.getMessage());
		}
		System.out.println("Thread3 Exit");
	}
}
public class ThreadExtending
{
	public static void main(String args[])
	{
		Thread t1=new Thread(new Thread1(),"ECE");
		Thread t2=new Thread(new Thread2(),"MEC");
```

```
		Thread t3=new Thread(new Thread3(),"EEE");
		t1.start();
		t2.start();
		t3.start();
	}
}
/*
D:\OpenElectiveOOP>java ThreadExtending
Hello
Welcome
Good Morning
Good Morning
Hello
Good Morning
Welcome
Good Morning
Hello
Good Morning
Thread1 exit
Hello
Welcome
Hello
Welcome
Thread2 exit
Welcome
Thread3 Exit*/
```

//java program to create thread by implementing runnable interface

```
import java.lang.*;
class Thread1 implements Runnable
{
	public void run()
	{
		try
		{
			for(int i=0;i<5;i++)
			{
				System.out.println("Good Morning");
				Thread.sleep(1000);
			}
```

```
		}
		catch(InterruptedException e)
		{
			System.out.println(e.getMessage());
		}
		System.out.println("Thread1 exit");
	}
}
class Thread2 implements Runnable
{
	public void run()
	{
		try
		{
			for(int i=0;i<5;i++)
			{
				System.out.println("Hello");
				Thread.sleep(2000);
			}
		}
		catch(InterruptedException e)
		{
			System.out.println(e.getMessage());
		}
		System.out.println("Thread2 exit");
	}
}
class Thread3 implements Runnable
{
	public void run()
	{
		try
		{
			for(int i=0;i<5;i++)
			{
				System.out.println("Welcome");
				Thread.sleep(3000);
			}
		}
		catch(InterruptedException e)
		{
```

```
                System.out.println(e.getMessage());
            }
            System.out.println("Thread3 Exit");
        }
}
public class ThreadImplementing
{
        public static void main(String args[])
        {
            Thread t1=new Thread(new Thread1(),"ECE");
            Thread t2=new Thread(new Thread2(),"MEC");
            Thread t3=new Thread(new Thread3(),"EEE");
            t1.start();
            t2.start();
            t3.start();
        }
}
/*D:\OpenElectiveOOP>java ThreadImplementing
Hello
Welcome
Good Morning
Good Morning
Hello
Good Morning
Welcome
Good Morning
Hello
Good Morning
Thread1 exit
Hello
Welcome
Hello
Welcome
Thread2 exit
Welcome
Thread3 Exit*/
```

```java
//java program to implement thread priorities
import java.lang.*;
class Thread1 extends Thread
{
	public void run()
	{
		try
		{
			System.out.println("ECE");
			Thread.sleep(1000);
		}
		catch(InterruptedException e)
		{
			System.out.println(e.getMessage());
		}
		System.out.println("Thread1 Exit");
	}
}
class Thread2 extends Thread
{
	public void run()
	{
		try
		{
			System.out.println("MECH");
			Thread.sleep(2000);
		}
		catch(InterruptedException e)
		{
			System.out.println(e.getMessage());
		}
		System.out.println("Thread2 Exit");
	}
}
class Thread3 extends Thread
{
	public void run()
	{
		try
		{
			System.out.println("CIV");
```

```
                Thread.sleep(3000);
            }
            catch(InterruptedException e)
            {
                System.out.println(e.getMessage());
            }
            System.out.println("Thread3 Exit");
        }
}
public class ThreadPriority
{
        public static void main(String args[])
        {
            Thread t1=new Thread(new Thread1());
            Thread t2=new Thread(new Thread2());
            Thread t3=new Thread(new Thread3());
            t1.setPriority(1);
            t2.setPriority(Thread.NORM_PRIORITY);
            t3.setPriority(7);
            t1.start();
            t2.start();
            t3.start();
        }
}
/*
Output
D:\OpenElectiveOOP>java ThreadPriority
MECH
CIV
ECE
Thread1 Exit
Thread2 Exit
Thread3 Exit*/
```